Michele Bilko

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EDUCATION

Boston University, Boston, MA Sep 2024 – May 2026
B.A. in Computer Science, Minor in Philosophy · Dean's List · Graduated 05/26
Relevant Coursework: Image & Video Computing (Computer Vision), Advanced Algorithms, Fine-Grained Complexity, Theory of Computation, Probability in Computing, Database Systems

Tufts University, Medford, MA Sep 2026 – May 2028
M.S. in Computer Science · Incoming Fall 2026 · Expected Graduation: 05/28

University of California, Santa Cruz, CA Sep 2023 – Jun 2024
B.S. coursework in Computer Science (transferred) · Dean's Honors List
Relevant Coursework: Machine Learning, Data Structures & Algorithms, Computer Systems & C Programming

RESEARCH EXPERIENCE

Directed Study in Graph Algorithms — Boston University Jan 2026 – Present
Advised by Prof. Dora Erdős

- Investigating graph conductance and open problems in network connectivity, spanning theoretical foundations and algorithmic applications in algorithmic graph theory.
- Implemented published algorithm pseudocode and built a Python/NetworkX framework benchmarking distributed conductance-testing algorithms (BTT vs. Fichtenberger–Vasudev) across varied graph topologies.

Satellite Image Restoration Research — Boston University Jan 2026 – May 2026
Computer Vision / Deep Learning

- Developed a deep-learning pipeline that reconstructed all of the ~22% stratified data gaps in Landsat 7 imagery caused by the 2003 Scan Line Corrector failure.
- Extended the LaMa (Large Mask Inpainting) model with multi-temporal reference-image guidance and CBAM attention mechanisms to preserve physical terrain features across timestamps.

PUBLICATIONS & RESEARCH WRITING

- *Deep-learning inpainting for Landsat 7 SLC-off gap reconstruction* — paper in preparation (extends LaMa with multi-temporal guidance and CBAM attention).
- *Empirical comparison of distributed graph conductance-testing algorithms (BTT vs. Fichtenberger–Vasudev)* — paper in preparation, advised by Prof. Dora Erdős.
- Authored the Wikipedia article on Fine-Grained Complexity, synthesizing core results in the field for a broad technical audience.
- Wrote a graduate-level survey paper on fine-grained complexity (CS 530).

PROFESSIONAL EXPERIENCE

Research Intern — Columbia University Media Center for Art History Sep 2022 – Jun 2023

- Contributed to the design and ongoing maintenance of the Media Center's public-facing website: learn.columbia.edu/media-center-art-history.
- Engineered an automated data-processing pipeline (Java, Python, pandas) transforming 20,000+ lines of imaging metadata into a normalized, searchable schema for digital-humanities research infrastructure.
- Applied photogrammetry and 3D-reconstruction algorithms using Structure-from-Motion (SfM) pipelines to build virtual tours.

COMMUNITY & VOLUNTEER PROJECTS

Central Rock Gym Route Tracking System · Django, React, PostgreSQL Aug 2025 – Present

- Built a full-stack web application with a Django backend (custom ORM queries and database triggers) and React frontend, integrating real-time data synchronization and a D3.js analytics dashboard over a production SQL database.

Boston Music Zine Website · Freelance Web Development 2026 – Present

- Designing and building a website for a local Boston house-show music zine.

TECHNICAL SKILLS

Programming Languages: Python, C++, SQL, Java, C#, JavaScript/TypeScript
ML & Data: PyTorch, TensorFlow, scikit-learn, NumPy, pandas
Tools & Libraries: SQLAlchemy, Web Scraping (requests, BeautifulSoup, Selenium), NetworkX, LaTeX, Git, Linux/Unix
Research Areas: Computer Vision, Deep Learning, Machine Learning, Graph Algorithms, Algorithmic Optimization, Computational Complexity
AI Tools: Productive with LLM-based coding assistants and agents for prototyping and task automation.